VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

FACULTY : SWETA BHATTACHARYA

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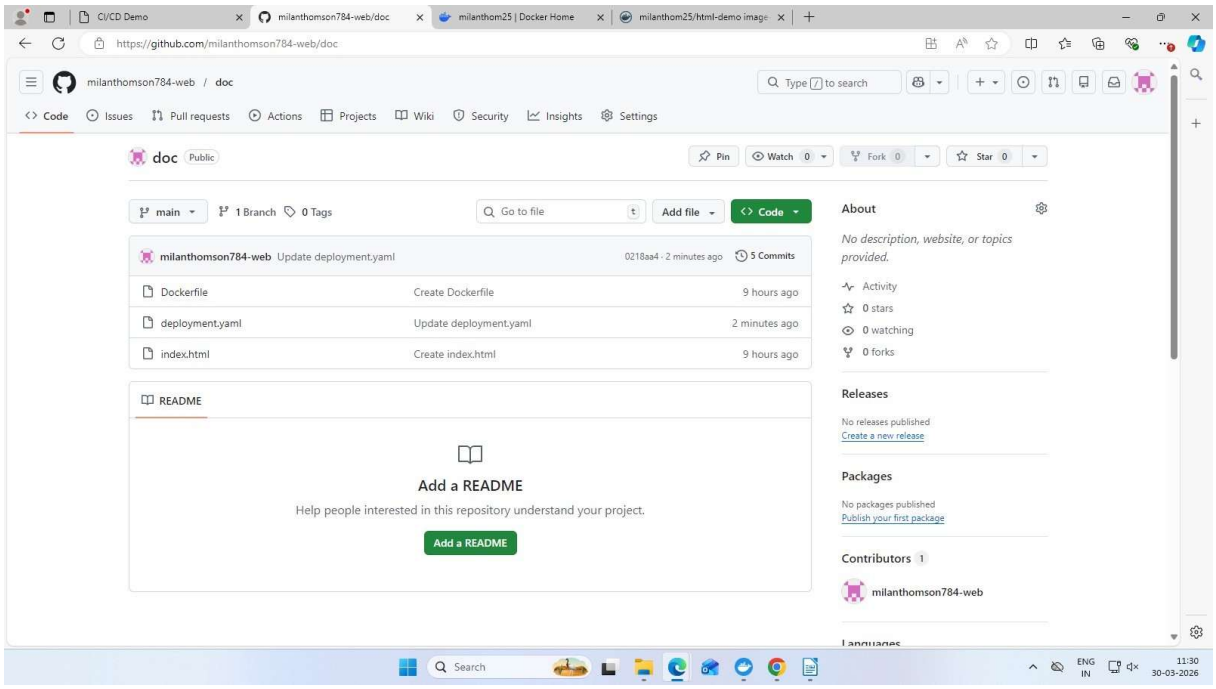
SUBJECT : AGILE DEVELOPMENT PROCESS AND DEVOPS

COURSE CODE: ISWE406P

1. Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile**
- **Deployment.yaml**



2. Enable Kubernetes

• Open the Docker Dashboard: Launch the Docker Desktop application on your system.

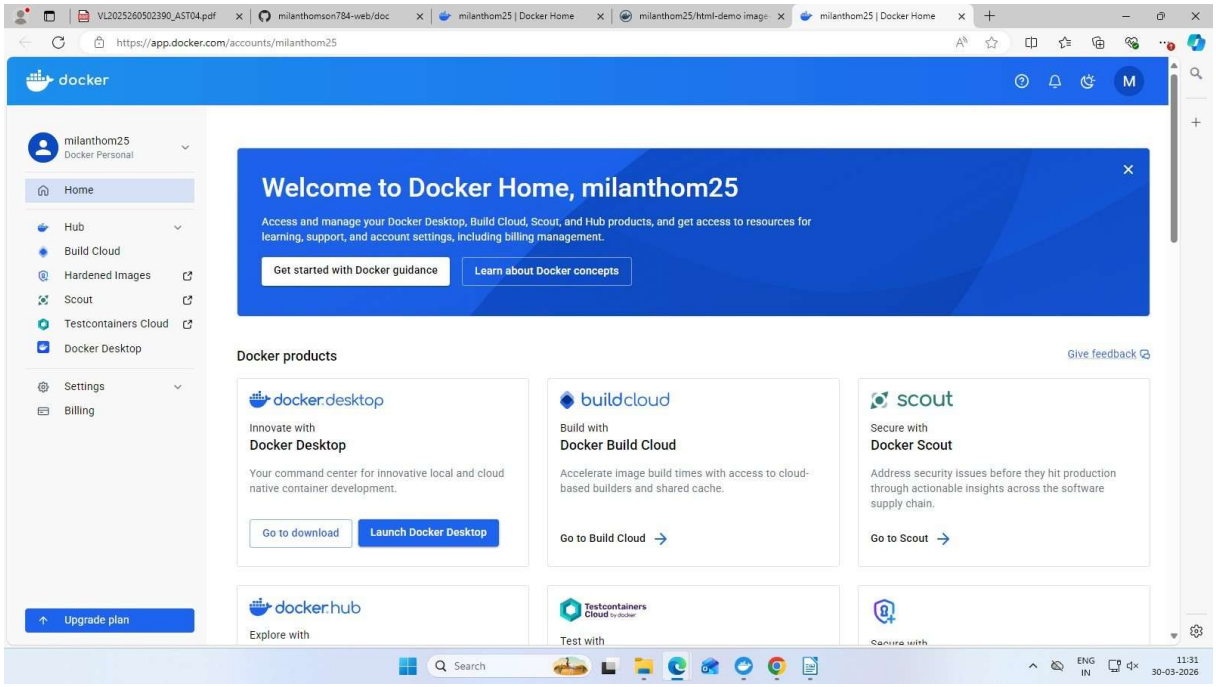
• Navigate to Settings: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.

• Select the Kubernetes Tab: In the Settings menu, find and select the Kubernetes tab.

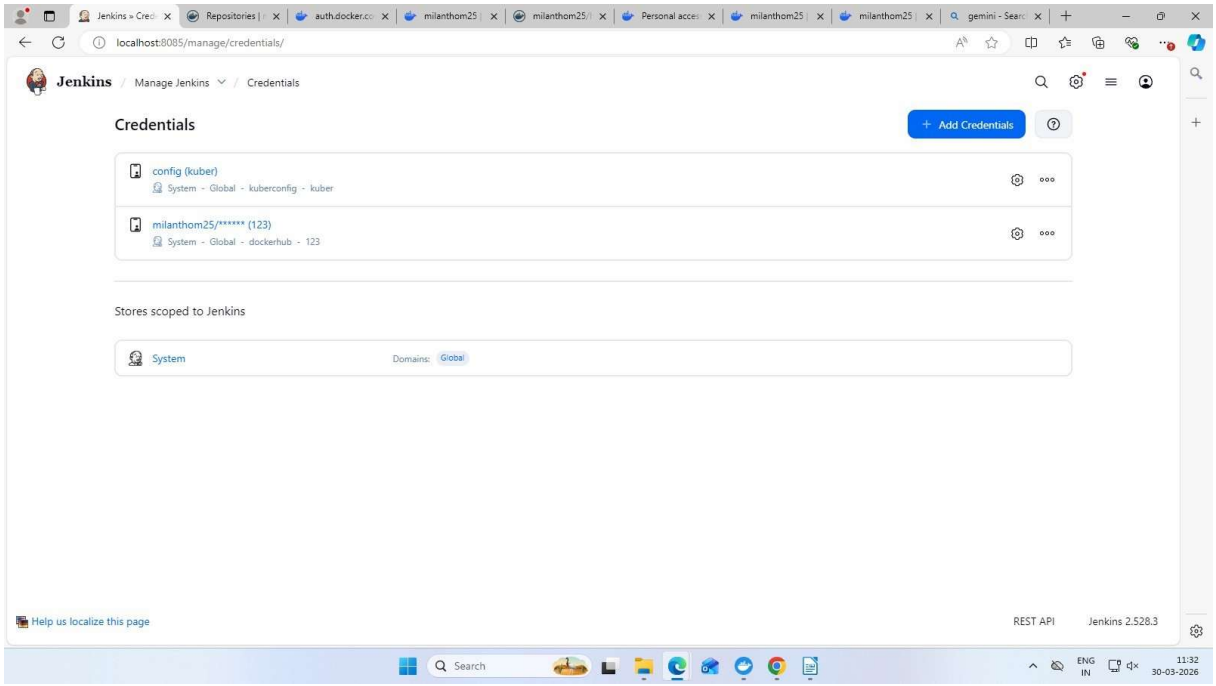
• Enable Kubernetes: Check the box labeled Enable Kubernetes.

• Apply & Restart: Click the Apply & Restart button at the bottom of the settings window.

Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time

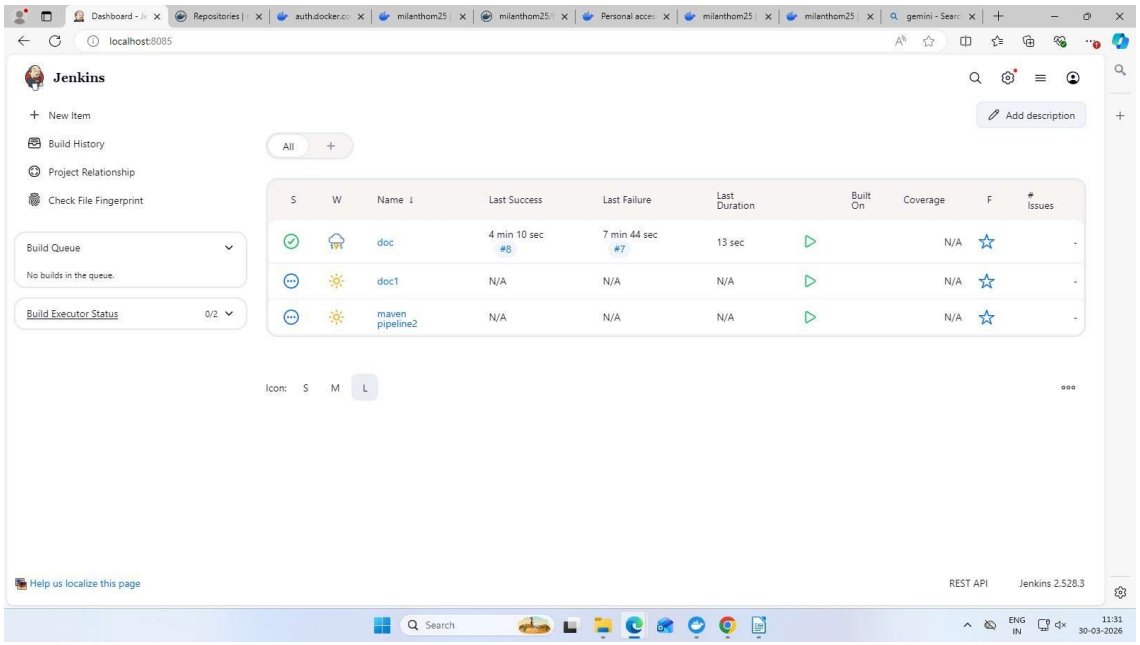


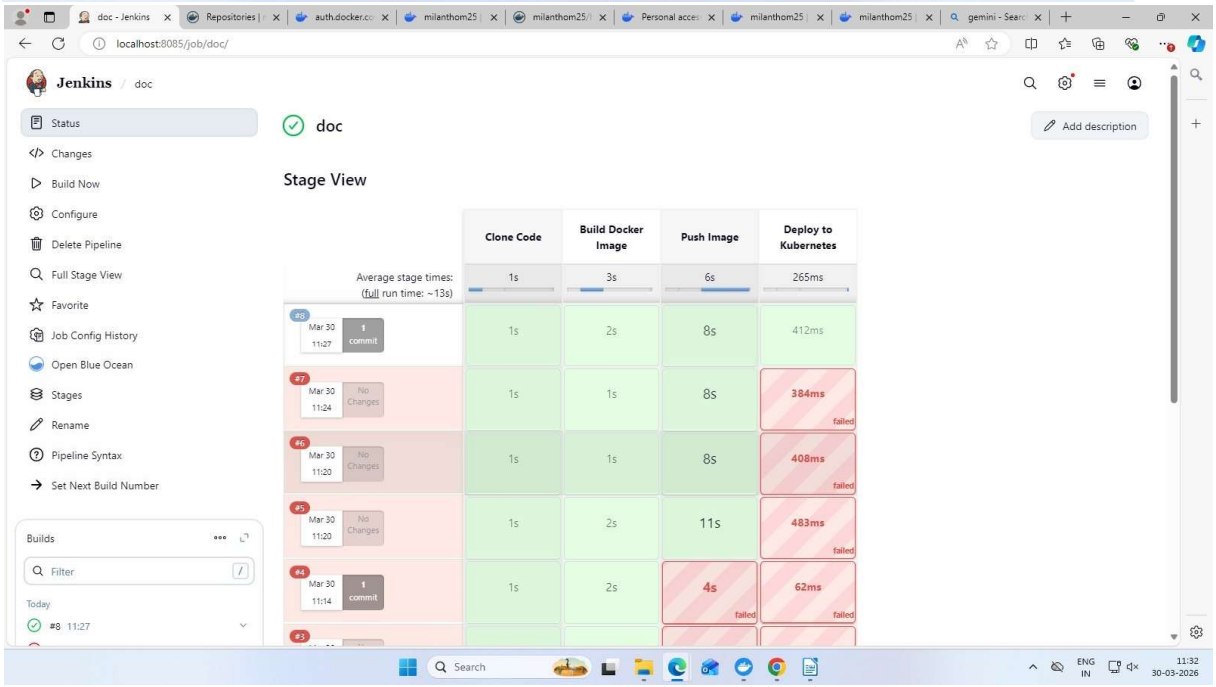
3. Open Jenkins
In Jenkins Setting --> Credentials
New Credentials – User name: <should use the Docker Desktop username>, password <same as Docker Desktop password>, ID:<dockerhub>, Description <any>

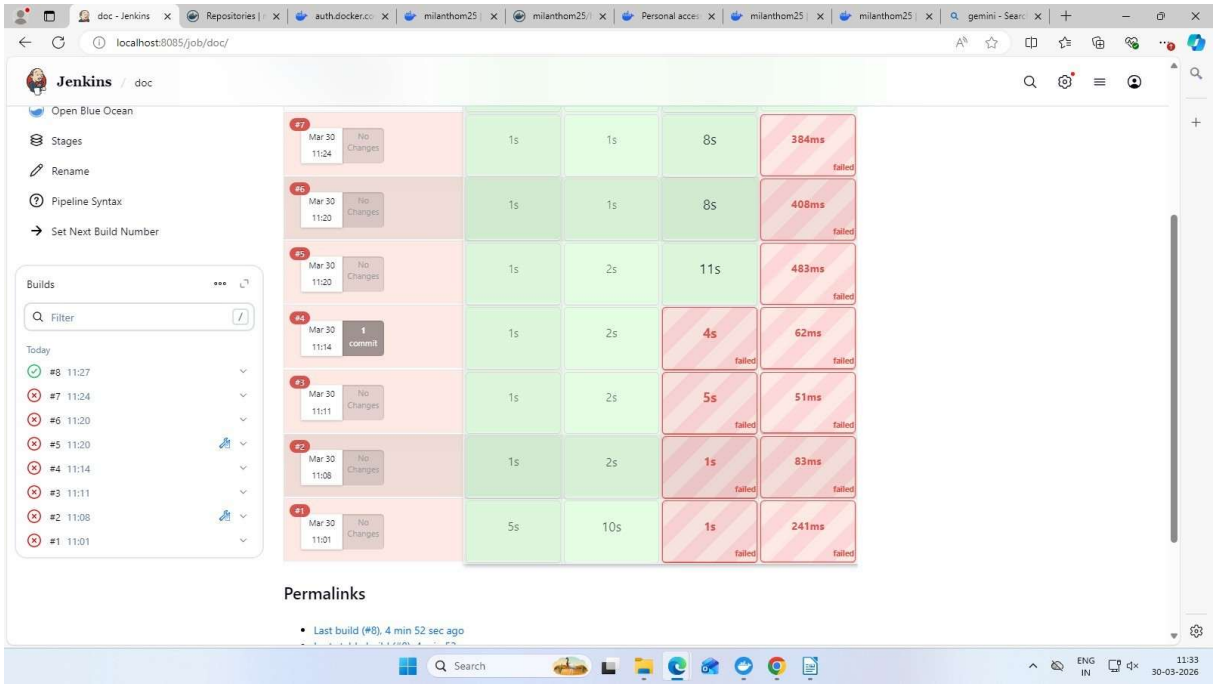


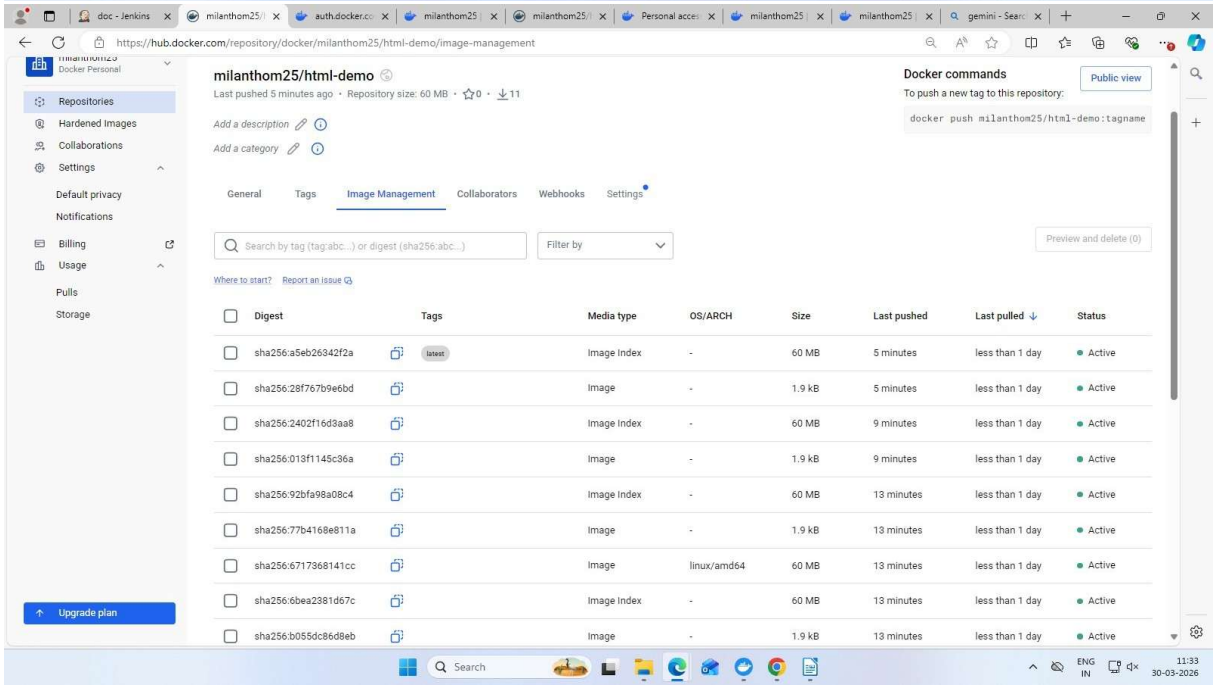
Create New Item -> Pipeline
Paste the code in pipeline script
- Update URL of GIT repository
- Make sure credentialsId: “dockerhub” same as mentioned in credentials

- **Make sure that the ID mentioned in credentials "kuberconfig" is same as in script**
- **Run the Build**






doc - Jenkins
localhost:8085/job/doc/
Jenkins / doc
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Set Next Build Number
Builds
Filter
Today
#8 11:27
#7 11:24
#6 11:20
#5 11:20
#4 11:14
#3 11:11
#2 11:08
#1 11:01
#7 Mar 30 11:24 No Changes 1s 1s 8s 384ms failed
#6 Mar 30 11:20 No Changes 1s 1s 8s 408ms failed
#5 Mar 30 11:20 No Changes 1s 2s 11s 483ms failed
#4 Mar 30 11:14 1 commit 1s 2s 4s failed 62ms failed
#3 Mar 30 11:11 No Changes 1s 2s 5s failed 51ms failed
#2 Mar 30 11:08 No Changes 1s 2s 1s failed 83ms failed
#1 Mar 30 11:01 No Changes 5s 10s 1s failed 241ms failed
Permalinks
Last build (#8), 4 min 52 sec ago


https://hub.docker.com/repository/docker/milanthom25/html-demo/image-management
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade plan
milanthom25/html-demo
Last pushed 5 minutes ago • Repository size: 60 MB • 0 • 11
Add a description
Add a category
Docker commands
Public view
To push a new tag to this repository:
docker push milanthom25/html-demo:tagname
General
Tags
Image Management
Collaborators
Webhooks
Settings
Search by tag (tag:abc...) or digest (sha256:abc...)
Filter by
Preview and delete (0)
Where to start?
Report an issue
Digest
Tags
Media type
OS/ARCH
Size
Last pushed
Last pulled
Status
sha256:a5eb26342f2a latest Image Index - 60 MB 5 minutes less than 1 day Active
sha256:28f767b9e6bd Image - 1.9 kB 5 minutes less than 1 day Active
sha256:2402f16d3aa8 Image Index - 60 MB 9 minutes less than 1 day Active
sha256:013f1145c36a Image - 1.9 kB 9 minutes less than 1 day Active
sha256:92bfa98a08c4 Image Index - 60 MB 13 minutes less than 1 day Active
sha256:77b4168e811a Image - 1.9 kB 13 minutes less than 1 day Active
sha256:6717368141cc Image linux/amd64 60 MB 13 minutes less than 1 day Active
sha256:6bea2381d67c Image Index - 60 MB 13 minutes less than 1 day Active
sha256:b055dc86d8eb Image - 1.9 kB 13 minutes less than 1 day Active

https://hub.docker.com/repository/docker/milanthom25/html-demo/image-management

hub | Explore | My Hub | Search Docker Hub (CtrlK)

milanthom25
Docker Personal

- Repositories
- Hardened Images
- Collaborations
- Settings
  - Default privacy
  - Notifications
- Billing
- Usage
  - Pulls
  - Storage

Upgrade plan

Repositories / html-demo / Image-management

milanthom25/html-demo

Last pushed 5 minutes ago • Repository size: 60 MB • 0 • 11

Add a description
Add a category

Docker commands

Public view

To push a new tag to this repository:

```
docker push milanthom25/html-demo:tagname
```

General | Tags | Image Management | Collaborators | Webhooks | Settings

Search by tag (tag:abc...) or digest (sha256:abc...) | Filter by | Preview and delete (0)

Where to start? Report an issue

Digest	Tags	Media type	OS/ARCH	Size	Last pushed	Last pulled	Status
sha256:a5eb26342f2a	latest	Image Index	-	60 MB	5 minutes	less than 1 day	Active
sha256:28f767b9e6bd		Image	-	1.9 kB	5 minutes	less than 1 day	Active
sha256:2402f16d3aa8		Image Index	-	60 MB	9 minutes	less than 1 day	Active
sha256:013f1145c36a		Image	-	1.9 kB	9 minutes	less than 1 day	Active
sha256:92bfa98a08c4		Image Index	-	60 MB	13 minutes	less than 1 day	Active
sha256:77b4168e811a		Image	-	1.9 kB	13 minutes	less than 1 day	Active
sha256:6717368141cc		Image	linux/amd64	60 MB	13 minutes	less than 1 day	Active

Search | ENG IN | 11:33 30-03-2026

https://hub.docker.com/repository/docker/milanthom25/html-demo/general

hub | Explore | My Hub | Search Docker Hub (CtrlK)

milanthom25
Docker Personal

- Repositories
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- Billing
- Usage
  - Pulls
  - Storage

Upgrade plan

Repositories / html-demo / General

milanthom25/html-demo

Last pushed 6 minutes ago • Repository size: 60 MB • 0 • 13

Add a description
Add a category

Docker commands

Public view

To push a new tag to this repository:

```
docker push milanthom25/html-demo:tagname
```

General | Tags | Image Management | Collaborators | Webhooks | Settings

Tags

DOCKER SCOUT INACTIVE
Activate

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	6 minutes

See all

Repository overview

INCOMPLETE

An overview describes what your image does and how to run it. It displays in the public view of your repository once you have pushed some content.

Add overview

buildcloud

Build with
Docker Build Cloud

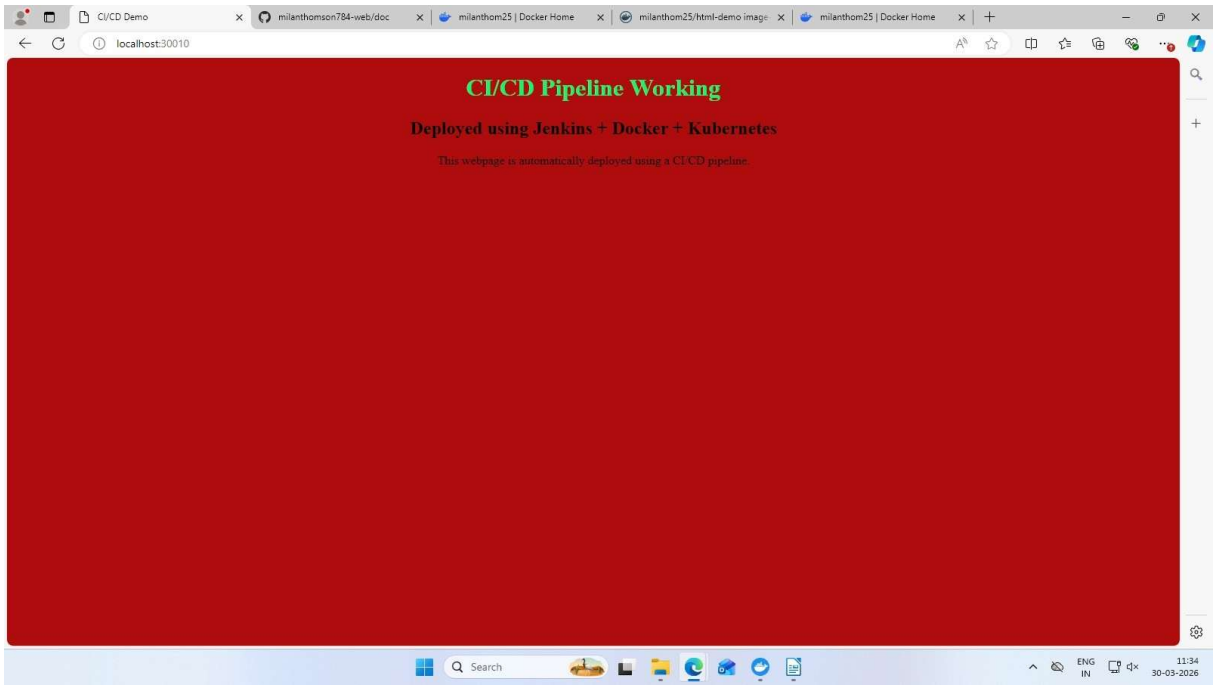
Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

Go to Docker Build Cloud →

https://hub.docker.com/repository/docker/milanthom25/html-demo/general

Search | ENG IN | 11:34 30-03-2026

Also the HTML page should open in the port mentioned in deployment.yaml file:
nodePort: 30010
So check if the HTML file open in http://localhost:30010